

Question

**Write a Java program to demonstrate Multilevel Inheritance using constructors, method overriding and method overloading.
Calculate the area of a rectangle.**

Steps

1. Create a class `Shape`.
2. Declare length and breadth.
3. Create a constructor to initialize them.
4. Create a `calculate()` method.
5. Create a class `Rectangle` extending `Shape`.
6. Create a constructor in `Rectangle`.
7. Create another class `Area` extending `Rectangle`.
8. Create a constructor in `Area`.
9. Override the `calculate()` method.
10. Calculate area using `length × breadth`.
11. Overload the `calculate()` method.
12. Create an object of `Area` in `main()`.
13. Call the methods and display the area.

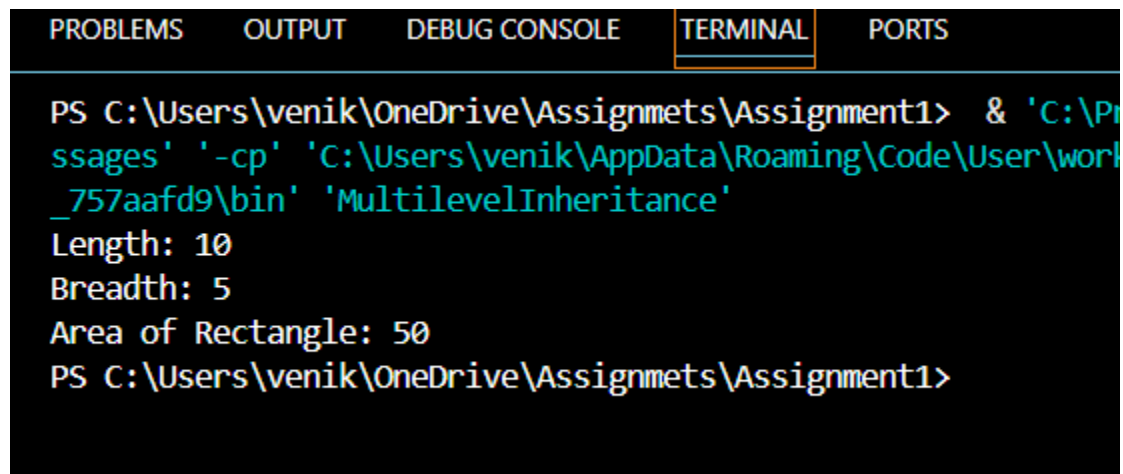
Algorithm

START

1. Create class `Shape`.
2. Declare length and breadth.
3. Initialize them using constructor.
4. Define `calculate()` method.
5. Create class `Rectangle` extending `Shape`.
6. Create `Rectangle` constructor.

7. Create class Area extending Rectangle.
8. Create Area constructor.
9. Override calculate().
10. Calculate Area = Length × Breadth.
11. Overload calculate() with different parameters.
12. Create Area object.
13. Call the methods.
14. Display the area.

STOP



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\venik\OneDrive\Assignmets\Assignment1> & 'C:\Program Files\Microsoft Visual Studio\2019\Community\MSBuild\Tools\MSBuild.exe' -cp 'C:\Users\venik\AppData\Roaming\Code\User\workspace\757aafd9\bin' 'MultilevelInheritance'
Length: 10
Breadth: 5
Area of Rectangle: 50
PS C:\Users\venik\OneDrive\Assignmets\Assignment1>
```